

DK14292/Rennecke

Title: Dissecting the moat regime at low energies: Renormalization and the phase structure

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We thank the referee for the very useful comments and for the positive assessment regarding its publication. In the following, we respond to each of the referee's five questions and comments. The manuscript has been revised accordingly. We hope that our responses satisfactorily address the referee's concerns.

Answer to Referee:

- **Comment 1:** *“While the moat regime is briefly defined in the introduction, followed by references to works with more detailed description, it would greatly improve comprehension if the introduction of the moat regime was expanded, including the characteristic properties of the systems in the moat regime, possible criteria for its detection and differences between the moat regime and inhomogeneous phase (mentioned later in the text) ”*

Answer: We agree with the referee that the description is a bit short. Although the general and precise definition of a moat regime, “...where the static (zero frequency) energy of mesons develops a minimum at nonzero momentum...”, was given in the second paragraph, we have expanded this a bit, also adding references [32, 33] for examples of observable signals of inhomogeneous phases. We still chose to refrain from explicit equations in the introductions, as they are given in the main text in great detail.

- **Comment 2:** *“The moat regime is identified as a region where the transverse pion wave function renormalization factor (Eq. 9) becomes negative - thus making $p=0$ a local maximum of the pion two-point function which should be followed by a minimum at some nonzero p . Is there any reason that prohibits existence of several local minima of the pion two-point function - with a global minimum at $p > 0$, and a local minimum at 0? This, in my understanding, would create a moat regime, while the transverse pion wave function renormalization factor remains positive. ”*

Answer: No, we do not see any physical reason to prohibit this behavior. That is why we also always check the entire momentum dependence of the two-point function, cf. Fig. 3. As far as we can tell, $Z^\perp(0) < 0$ is a sufficient, but not necessary criterion for the moat regime. However, we have not encountered the situation the referee describes in any of our calculations yet. This is related to the structure of the pion self energy. The wave function renormalization at finite momentum can be defined by

$$Z_\pi^\perp(\vec{p}) = \frac{\Sigma_\pi(\vec{p}) - \Sigma_\pi(\vec{p}=0)}{\vec{p}^2}. \quad (1)$$

So from Eq.(7) and Eq.(10) we can see $Z_\pi^\perp(\vec{p})$ is roughly

$$Z_\pi^\perp(\vec{p}) \approx 1 + h^2 N_c \int \frac{d^3 q}{(2\pi)^3} \mathcal{F} \mathcal{F}^-(q, p). \quad (2)$$

The threshold function $\mathcal{FF}^-(q, p)$ is suppressed at finite momentum, and $Z_\pi^\perp(\vec{p})$ monotonically returns to 1 as $\vec{p}$ increases. In our case, this starts to occur at fairly low momenta, so $Z_\pi^\perp(\vec{p})$ does not cross zero more than once. Therefore, at least in calculations where only the quark-loop contribution to the self-energy is taken into account, the only way to realize a moat regime is that the wave function renormalization is already negative at zero momentum. This is also true in the fRG computation in arXiv:2412.15949.

- **Comment 3:** *“The authors have shown that their renormalization scheme has no scale dependency, and additionally suggest a possibility of alternative conditions fixing the transverse renormalization - could they explain if such conditions can be proved to result in the same moat regime region making it truly renormalization scheme independent?”*

Answer: Unfortunately, providing such a proof is essentially impossible. Ideally, renormalization group invariance together with appropriate renormalization conditions should remove any scheme dependence. But in practice, this hardly ever happens (one-loop perturbation theory and anomalies are rare exceptions). In general, independence on the renormalization scheme is only possible in the absence of approximations or computations to all loop orders. Furthermore, different renormalization conditions will always lead to different predictions unless they are physically equivalent (in which case they should be viewed as the same condition). This is clear, as the renormalization conditions connect the regularized expressions, involving arbitrary counter terms, to physical quantities.

In this context, we realized that the sentences in the third paragraph of the summary, “In particular since certain choices for $Z^\perp$ might obscure the identification of chiral partners in the chirally restored phase. We will get back to this in a forthcoming work [71].”, can be confusing without further explanations. As this will be done in a forthcoming paper, we just dropped this part here. As we suggest above Eq. (24), it is possible to directly fix $Z^\perp$ from lattice data. We now clarify that this could be advantageous for model studies, as the relevant quantity is fixed directly.

- **Comment 4:** *“The paragraph around Eq. 28 tells that the moat regime appears in Random Phase Approximation for large T and arbitrary μ , including $\mu = 0$, and that this behavior disappears once authors take into account renormalizations of the Yukawa coupling - possible authors could add a sentence there explaining why this behavior is not expected in full theory ”*

Answer: In Sec. IV B, we have added several sentences to clarify the impact of the Yukawa coupling on the moat regime within a more complete QCD theory. We hope that this clarifies that the information encoded in the Yukawa coupling plays a significant role in determining the presence of the moat regime, in particular its disappearance at low chemical potential.

- **Comment 5:** *“There are couple cases of unclear English language in the Summary - the sentence “In particular since certain choices for $Z^{\text{transverse}}$ might obscure the identification of chiral partners in the chirally restored phase” seems to lack the main part (or needs to be joined with the previous one). “crucially depends to the scalar meson mass” should possibly be “crucially depends on the scalar meson mass”. I recommend fixing these issues, and making a thorough rereading of the paper to catch other possible problems. ”*

Answer: We fully agree, see our reply to Comment 3. We have also corrected “to” to “on”.

Other changes: We replaced the word “dispersion” by “static energy” as this is more accurate. Dispersions describe the frequencies at which the propagators have a pole at a given spatial momentum. The moat regime appears, as we describe in detail in the text, in the static energy at zero frequency. Fortunately, this imprecise language was used only in the introduction.